

HERMANUS, South Africa

WMO: 689180

Lat: 34.4325S

Lon: 19.2247E

Elev: 52

StdP: 14.67

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>7      | (b)<br>47.4 | (c)<br>48.9 | (d)<br>37.5                   | (e)<br>33.0 | (f)<br>58.4 | (g)<br>40.0 | (h)<br>36.4 | (i)<br>55.1 | (j)<br>25.4           | (k)<br>60.6 | (l)<br>23.3 | (m)<br>59.1 | (n)<br>4.1            | (o)<br>70 | (p)<br>0.401 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>1      | (b)<br>8.7             | (c)<br>79.2     | (d)<br>66.0 | (e)<br>76.6 | (f)<br>65.8 | (g)<br>74.7 | (h)<br>65.6 | (i)<br>69.6         | (j)<br>74.5 | (k)<br>68.5 | (l)<br>73.4 | (m)<br>67.5 | (n)<br>72.4 | (o)<br>9.3           | (p)<br>110 |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 67.9                            | 102.8 | 72.2 | 66.7 | 98.4 | 71.1 | 65.7 | 94.9 | 70.3 | 33.7          | 74.6 | 32.8 | 73.7 | 32.0 | 72.5 | 75.7              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |
| Min               |      | Max  |                            | Min  | Max  | Min                | Max                                                | Min       | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 23.4              | 20.9 | 18.7 |                            | 45.0 | 85.6 | 1.1                | 2.6                                                | 44.2      | 87.5 | 43.5       | 89.0 | 42.9       | 90.5 | 42.1       | 92.4 |
|                   |      |      |                            | DB   | WB   | 42.8               | 70.7                                               | 1.5       | 1.4  | 41.7       | 71.8 | 40.8       | 72.6 | 39.9       | 73.4 |
|                   |      |      |                            |      |      | 38.8               | 74.4                                               |           |      |            |      |            |      |            |      |

Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 63.0   | 68.4  | 68.2  | 66.6  | 64.0  | 61.2  | 58.4  | 57.4  | 57.7  | 59.5  | 62.4  | 64.6  | 67.4 |
|                                                                                  | DBStd       | 5.15   | 2.85  | 2.62  | 3.50  | 3.59  | 3.41  | 3.31  | 3.71  | 3.53  | 3.45  | 3.56  | 3.06  | 2.78 |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD65       | 1193   | 4     | 3     | 21    | 59    | 126   | 199   | 238   | 228   | 169   | 96    | 42    | 8    |
|                                                                                  | CDD50       | 4733   | 570   | 510   | 514   | 420   | 348   | 253   | 231   | 239   | 285   | 386   | 437   | 540  |
|                                                                                  | CDD65       | 451    | 109   | 94    | 70    | 29    | 8     | 3     | 4     | 2     | 4     | 16    | 29    | 83   |
|                                                                                  | CDH74       | 602    | 131   | 78    | 99    | 63    | 15    | 15    | 28    | 11    | 11    | 37    | 34    | 80   |
|                                                                                  | CDH80       | 61     | 12    | 4     | 12    | 9     | 2     | 3     | 6     | 1     | 1     | 5     | 2     | 4    |
| Wind                                                                             | WSAvg       | 8.1    | 8.4   | 7.8   | 7.9   | 7.0   | 6.9   | 7.7   | 7.6   | 8.2   | 8.6   | 9.2   | 9.2   | 8.6  |
| Precipitation                                                                    | PrecAvg     | 24.9   | 0.9   | 0.9   | 1.0   | 2.2   | 2.8   | 3.7   | 3.6   | 3.1   | 2.0   | 1.9   | 1.7   | 1.1  |
|                                                                                  | PrecMax     | 37.1   | 4.1   | 2.7   | 4.0   | 7.0   | 5.7   | 8.1   | 6.6   | 9.4   | 4.9   | 5.6   | 5.1   | 3.7  |
|                                                                                  | PrecMin     | 13.5   | 0.2   | 0.0   | 0.2   | 0.5   | 0.4   | 0.7   | 0.5   | 0.9   | 0.3   | 0.2   | 0.1   | 0.2  |
|                                                                                  | PrecStd     | 4.6    | 0.9   | 0.6   | 0.8   | 1.6   | 1.1   | 1.7   | 1.7   | 1.7   | 1.1   | 1.3   | 1.5   | 0.8  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 81.0  | 79.8  | 81.3  | 80.1  | 76.4  | 75.4  | 77.2  | 75.7  | 74.6  | 79.3  | 78.2  | 79.6 |
|                                                                                  |             | MCWB   | 68.3  | 67.5  | 67.3  | 63.7  | 59.6  | 57.1  | 56.8  | 56.3  | 61.2  | 65.8  | 64.6  | 67.4 |
|                                                                                  | 2%          | DB     | 77.0  | 76.4  | 76.5  | 74.6  | 70.7  | 68.3  | 70.0  | 67.9  | 69.5  | 73.2  | 74.0  | 76.2 |
|                                                                                  |             | MCWB   | 67.8  | 67.0  | 66.2  | 63.6  | 60.4  | 56.8  | 55.8  | 57.1  | 60.0  | 63.7  | 63.6  | 66.4 |
|                                                                                  | 5%          | DB     | 74.9  | 74.3  | 73.6  | 71.1  | 68.4  | 65.4  | 65.2  | 64.9  | 66.5  | 70.2  | 71.4  | 74.0 |
|                                                                                  |             | MCWB   | 67.0  | 66.6  | 65.7  | 63.0  | 59.9  | 57.1  | 56.7  | 56.9  | 58.9  | 62.0  | 63.5  | 65.6 |
|                                                                                  | 10%         | DB     | 73.2  | 72.8  | 71.6  | 68.9  | 66.5  | 63.6  | 63.0  | 62.8  | 64.5  | 67.8  | 69.6  | 72.4 |
|                                                                                  |             | MCWB   | 66.1  | 66.0  | 65.1  | 62.4  | 59.5  | 56.8  | 56.2  | 56.2  | 57.9  | 60.8  | 62.7  | 64.7 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 71.0  | 70.9  | 70.0  | 67.8  | 63.7  | 61.6  | 61.2  | 61.7  | 63.6  | 67.4  | 68.5  | 69.8 |
|                                                                                  |             | MCDB   | 75.9  | 74.5  | 75.5  | 72.9  | 68.1  | 65.7  | 68.1  | 66.7  | 68.6  | 74.7  | 72.9  | 75.8 |
|                                                                                  | 2%          | WB     | 69.6  | 69.2  | 68.6  | 66.1  | 62.6  | 60.0  | 59.1  | 60.0  | 61.7  | 65.0  | 66.4  | 68.0 |
|                                                                                  |             | MCDB   | 74.8  | 73.0  | 73.8  | 70.6  | 66.3  | 63.9  | 64.3  | 64.0  | 66.8  | 71.6  | 70.7  | 74.0 |
|                                                                                  | 5%          | WB     | 68.4  | 68.2  | 67.2  | 64.7  | 61.7  | 58.9  | 57.9  | 58.4  | 60.3  | 63.3  | 65.1  | 66.7 |
|                                                                                  |             | MCDB   | 73.3  | 72.3  | 72.2  | 69.1  | 65.7  | 62.9  | 63.0  | 63.1  | 65.1  | 68.8  | 69.4  | 72.3 |
|                                                                                  | 10%         | WB     | 67.1  | 67.1  | 66.1  | 63.5  | 60.8  | 57.8  | 56.7  | 56.9  | 58.9  | 62.0  | 63.7  | 65.7 |
|                                                                                  |             | MCDB   | 72.0  | 71.4  | 70.8  | 67.8  | 65.0  | 62.1  | 61.8  | 61.6  | 63.6  | 66.8  | 68.3  | 71.1 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.7   | 8.5   | 8.8   | 9.7   | 9.5   | 10.3  | 10.5  | 10.1  | 9.5   | 9.3   | 9.1   | 8.9  |
|                                                                                  |             | MCDBR  | 11.5  | 10.9  | 11.8  | 13.2  | 12.0  | 12.9  | 14.3  | 13.1  | 12.5  | 13.6  | 12.2  | 11.4 |
|                                                                                  | 5% WB       | MCWBR  | 5.4   | 5.3   | 5.4   | 6.0   | 6.1   | 6.8   | 7.2   | 6.6   | 6.6   | 6.7   | 6.1   | 5.9  |
|                                                                                  |             | MCWBR  | 9.6   | 8.3   | 10.2  | 10.9  | 9.7   | 9.9   | 12.2  | 11.2  | 11.6  | 11.9  | 9.6   | 10.1 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.352  | 0.354 | 0.344 | 0.328 | 0.323 | 0.316 | 0.318 | 0.335 | 0.347 | 0.360 | 0.347 | 0.352 |      |
|                                                                                  | taud        | 2.523  | 2.527 | 2.551 | 2.587 | 2.585 | 2.578 | 2.565 | 2.510 | 2.453 | 2.434 | 2.510 | 2.514 |      |
|                                                                                  | Ebn at Noon | 312    | 305   | 297   | 284   | 267   | 261   | 266   | 277   | 290   | 299   | 312   | 313   |      |
|                                                                                  | Edh at Noon | 35     | 34    | 31    | 26    | 23    | 22    | 23    | 28    | 33    | 37    | 36    | 36    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2470   | 2196  | 1762  | 1296  | 935   | 786   | 870   | 1115  | 1528  | 1962  | 2334  | 2511  |      |
|                                                                                  | RadStd      | 74     | 76    | 89    | 60    | 50    | 43    | 42    | 52    | 69    | 101   | 87    | 94    |      |

Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (12 neighbors) | +0.65 | +0.88   | N/A    | N/A     | +0.95 | +0.85 | -4          | -133  | +200  | +110  |

Nomenclature: See separate page